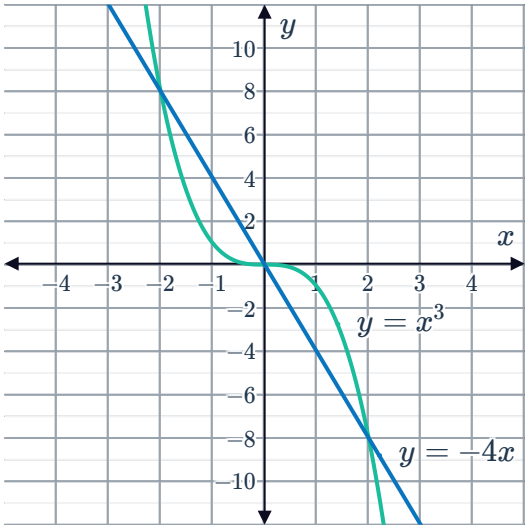
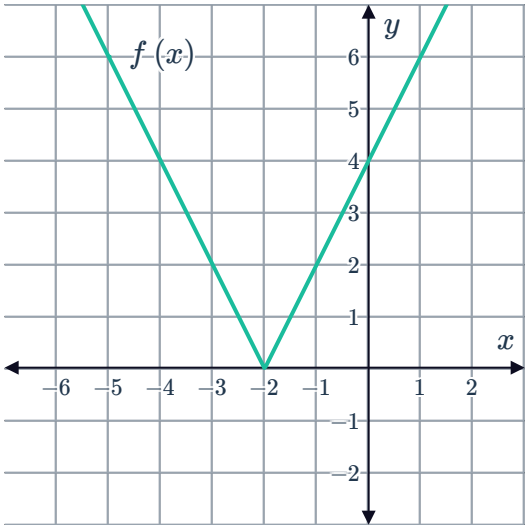
Solve equations graphically

1 The functions $y = -x^3$ and $y = -4x$ are shown on the following graph:
How many values of x satisfy the equation $-x^3 = -4x$?



2 Consider the following graph of the function $f(x) = |2x + 4|$:

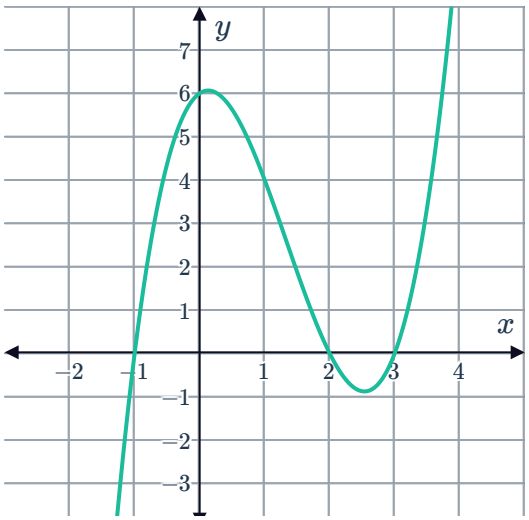


a Sketch the graph of $g(x) = 2$ on the same coordinate plane as $f(x)$.

b State the number of solutions for the equation $|2x + 4| = 2$.

c Find the solutions to the equation $|2x + 4| = 2$.

3 Consider the graph of the cubic function $y = (x + 1)(x - 2)(x - 3)$:



a State the number of solutions to the equation $(x + 1)(x - 2)(x - 3) = 0$.

b Determine the solutions to the equation $(x + 1)(x - 2)(x - 3) = 0$.

c State the number of solutions to the equation $(x + 1)(x - 2)(x - 3) = -2$.

d State the number of solutions to the equation $(x + 1)(x - 2)(x - 3) = 6$.

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4 Consider the functions $y = 9$ and $y = x^2$.

- **a** Sketch the graph of the functions on the same coordinate plane.
- **b** State the coordinates of the points of intersection.
- **c** Now, state the solutions to the equation $x^2 = 9$.

5 Consider the quadratic function $y = x^2 + 4$ and the linear function $x + y = 4$.

- **a** Sketch the graph of the functions on the same coordinate plane.
- **b** Now, state the number of solutions to the equation $x^2 + 4 = -x + 4$.
- **c** Determine the coordinates of the points of intersection of the two functions.

6 Consider the quadratic function $y = x^2 - 3$ and the linear function $x + y = 3$.

- **a** Sketch the graph of the functions on the same coordinate plane.
- **b** Now, state the number of solutions to the equation $x^2 - 3 = -x + 3$.
- **c** Determine the coordinates of the points of intersection of $y = x^2 - 3$ and $x + y = 3$.

7 Consider the functions $f(x) = -4x + x^2$ and $g(x) = -12 + 6x - x^2$.

- **a** Sketch the graph of $f(x)$ and $g(x)$ on the same coordinate plane.
- **b** Determine the coordinates of the points of intersection of $f(x)$ and $g(x)$.
- **c** Now, determine the solutions to the equation $-4x + x^2 = -12 + 6x - x^2$.

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Solve equations using technology

8 Use technology to solve the following equations. Round your answer to two decimal places where necessary.

a $x - 5 = \sqrt{x + 3}$

b $\sqrt[3]{x + 13} = 2$

c $x^{\frac{5}{3}} = 5 - 4x^2$

d $\sqrt{x + 4} + \sqrt{4x + 3} = 3$

e $\sqrt{3x - 2} = \sqrt{x} - \frac{1}{2}$

f $3^x = \frac{1}{x}$

g $2^{x+2} = 3^{x+3}$

h $2x - 5 = |x - 4|$

i $|x + 4| - 6 = -|x + 2| + 6$

j $\frac{3x^3}{x + 2} = \frac{4x^2}{x - 5}$

k $3x - 2 = 5x^3 + 2x^2 - x - 6$

l $3x^2 - 2x - 40 = -x^2 + 2x + 8$

m $2x^4 + 5x^3 + x = 3x^4 + 12x^3 - 81x^2 - 566x$

n $2x^3 + 3x^2 + 4x + 5 = 7x^3 + 6x^2 + 5x + 4$

9 Solve $3^{-x} = \sqrt{x + 4}$. Round your answer to three decimal places.

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Applications

10 Ned is looking to find approximate solutions to the simultaneous equations:

$$f(x) = -2^x + 9$$
$$g(x) = x^3 + 3x + 2$$

a Given that $h(x) = g(x) - f(x)$, write an expression for $h(x)$ in simplified form.

b Complete the following table of values for $h(x)$:

t	-2	-1	0	1	2
$h(x)$					

c The value of x that satisfies $f(x)$ and $g(x)$ simultaneously lies in which interval?

11 The populations, in thousands, of two competing species of animals is given by $f(t) = 215t + 3$ and $g(t) = \dfrac{1}{t-3}$, where $f(t)$ and $g(t)$ represent the respective populations t years from now. Jenny is looking to determine approximately when their populations will be equal.

a She first equates the two functions and forms a new quadratic equation $h(t)$. Write an equation for $h(t)$ in the form $at^2 + bt + c$, where $a > 0$.

b Complete the table of values, rounding the answer to two decimal places if necessary:

t	3	3.25	3.5	3.75	4
$h(t)$					

c Using the table from part (b), state what value of t should Jenny use to give an approximation to the point of intersection of $f(t)$ and $g(t)$.

d The equation $h(t) = 0$ has two solutions, but Jenny ignored the other solution. Explain why the other solution was ignored.

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